Logarithms

Name _____

Algebra 2 Lesson 6-3 (fast track) Date _____ Period _____

Learning goal: I can evaluate a logarithm by finding the exponent that gives the argument, and solve an equation by switching between logarithmic form and exponential form.

Language goal: I can explain how I evaluated a logarithm or solved an equation using the words base, exponent, argument, logarithmic form, and exponential form.

On the quiz: (1) Name the inverse of $y = b^x$: the box below (full Savvas quiz only). (2) Estimate ln of a number with a calculator (full Savvas quiz only). (3) Rewrite a logarithmic equation in exponential form: both Examples do this step. (4) Select the correct evaluations, and reject claims assigning a numerical value to an undefined logarithm: Example 3 and the Practice. (5) Solve an equation with a common logarithm in it: Example 5.

Logarithms. $\log_b y = x$ if and only if $b^x = y$ ($b > 0$, $b \neq 1$, $y > 0$). A logarithm *is* an exponent.

	Exponential form	**Logarithmic form**
Algebra	$b^x = y$	$\log_b y = x$
Words	the **base** raised to the **exponent** is the result	the log, base b, of the result (the **argument**) is the exponent
Numbers	$3^4 = 81$	$\log_3 81 = 4$

$\log x$ means base 10 (the **common logarithm**); $\ln x$ means base e (the **natural logarithm**).
$y = \log_b x$ is the inverse of $y = b^x$. Spanish: logaritmo, logaritmo común, función exponencial.

Example 3: Evaluate Logarithms

What is the value of each logarithmic expression? Ask: the base raised to *what exponent* gives the argument?

A. $\log_5 125$
THINK: $5^{?} = 125$
Since $5^3 = 125$, $\log_5 125 = 3$.

B. $\log_{\frac{1}{4}} 16$
THINK: $\left(\frac{1}{4}\right)^{?} = 16$
Since $\left(\frac{1}{4}\right)^{-2} = 4^2 = 16$, $\log_{\frac{1}{4}} 16 = -2$.

C. $\log_3 0$
THINK: $3^{?} = 0$
There is no such power, so $\log_3 0$ is **undefined**.

D. $\log_2 2^8$
THINK: $2^{?} = 2^8$
Since $2^8 = 2^8$, $\log_2 2^8 = 8$.

Why C is undefined: b^x is always positive when $b > 0$, so the argument of a logarithm must be positive. A logarithm itself *can* be negative (Part B).

Say it: To evaluate $\log_5 125$, I ask what exponent on the base _____ gives the argument _____. The exponent is _____, so $\log_5 125$ = _____. $\log_3 0$ is undefined because ...

Try It 3: Evaluate Logarithms

What is the value of each logarithmic expression?

Hint: Write the THINK step in exponential form first.

a. $\log_3\left(\frac{1}{81}\right)$
THINK: _____

b. $\log_7(-7)$
THINK: _____

c. $\log_5 5^9$
THINK: _____

Value: _____

Value: _____

Value: _____

Part a and Example 3 Part B both come out negative. Which part of the expression is a fraction in each one: the base or the argument?

Practice: Check the work

Each logarithm below has been evaluated. Circle the letter of **every** one that is correct. For each one that is wrong, write the exponential form that shows why (or write "undefined").

A. $\log 1 = 0$ _____

B. $\log_3 8 = 2$ _____

C. $\log \frac{1}{1000} = \frac{1}{3}$ _____

D. $\log_4(-1) = \frac{1}{4}$ _____

E. $\log_6 \frac{1}{36} = -2$ _____

F. $\ln e = 1$ _____

Which base does log have when no base is written? Which base does ln have?

Example 5: Solve Equations With Logarithms

What is the solution to each equation? Round to the nearest thousandth.

A. $25 = 10^{x-1}$

The variable is in the **exponent**. Convert to logarithmic form (base 10, exponent $x - 1$, result 25):

1. $25 = 10^{x-1}$
2. $\log 25 = x - 1$
3. $1 + \log 25 = x$
4. $2.398 \approx x$

Solution: $x \approx 2.398$.

Common error: 10 is not a coefficient, it is a base. You cannot divide both sides by 10 and then add 1.

B. $\ln(2x + 3) = 4$

The variable is in the **argument**. Convert to exponential form (base e, exponent 4, result $2x + 3$):

1. $\ln(2x + 3) = 4$
2. $2x + 3 = e^4$
3. $2x + 3 \approx 54.598$
4. $2x \approx 51.598$
5. $x \approx 25.799$

Solution: $x \approx 25.799$.

Check: $\ln\Big(2(25.799) + 3\Big) = \ln 54.598 \approx 4$. ✓

Say it: In Part B, I rewrote the logarithmic form $\ln(2x + 3) = 4$ in exponential form. The base is _____, the exponent is _____, and the argument $2x + 3$ equals _____. Then I . . .

Try It 5: Solve Equations With Logarithms

Solve each equation. Round to the nearest thousandth.

a. $\log(3x - 2) = 2$

Base _____ exponent _____ argument _____

Exponential form: _____

$x =$ _____

b. $e^{x+2} = 8$

Base _____ exponent _____ result _____

Logarithmic form: _____

$x \approx$ _____

Why did you switch to exponential form in part a but to logarithmic form in part b? Use the words *exponent* and *argument*.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
8RK ★	Convert between exponential and logarithmic form: all bases	Quiz 3; the first step of Quiz 5 (Examples 3 and 5)
GBR ★	Evaluate logarithms	Quiz 4 (Example 3, Practice)
BXU ★	Solve logarithmic equations I	Quiz 5 (Example 5, Try It 5a)
TDF	Evaluate logarithms using a calculator (optional)	Quiz 2 (full Savvas quiz only); the calculator step in Example 5
LWE	Evaluate exponential functions	Warm-up: powers like 5^3 and $\left(\frac{1}{4}\right)^{-2}$

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Use Structure

A student says that $\log_3\left(\frac{1}{27}\right)$ simplifies to -3. Is the student correct? Explain.

The student is _____ because the base _____ raised to the exponent _____ gives the argument _____.

Before the quiz, finish these sentences.

A logarithm is an _____: $\log_b y$ is the exponent you put on the base _____ to get the argument _____.

$\log_b 0$ and $\log_b(-1)$ are _____ because a positive base raised to any exponent is always _____.

To solve log(expression) $= c$, rewrite it in _____ form: $10^c =$ _____.